

Immersive Brain Exploration: A VR-Based Gamified System for Neuroanatomy Education with AI-Augmented Learning

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Figure 1: Overview of the VR neuroanatomy system: (a) Study Mode with interactive 3D brain exploration (b) Game Mode featuring spatial puzzle challenges

Abstract

This paper presents a VR system for neuroanatomy education targeting challenges in spatial complexity, knowledge integration, and long-term retention. The system combines immersive VR with AI-assisted visualizations and gamified interactions to encourage multimodal engagement with anatomical structures. A small formative probe suggests potential benefits for spatial understanding and motivation. Future work will conduct a systematic evaluation of educational effectiveness and explore lightweight personalization.

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1 Introduction

Neuroanatomy is difficult to master due to its complex three-dimensional organization and highly interconnected structures. Learners must reconcile multiple spatial perspectives and link morphology, vascular supply, functional pathways, and clinical syndromes—tasks that

demand strong spatial reasoning and durable knowledge retention [Roach et al. 2022]. While textbooks, atlases, and 2D videos remain valuable, their static and fragmented nature limits the formation of robust 3D mental models and cross-modal integration, motivating more immersive and interactive approaches [Alim et al. 2024].

Virtual Reality (VR) affords life-scale exploration, free manipulation, and direct experience of spatial arrangements [Adnan et al. 2025; Moro et al. 2021]. In parallel, generative-AI-assisted visuals can provide mnemonic and semantic cues that complement VR-based exploration, supporting visual-associative learning and recall. However, many existing VR systems lack coherent strategies to organize and contextualize such AI-assisted assets during study and task-based practice.

This paper presents a VR neuroanatomy system that integrates immersive 3D visualization with curated, AI-assisted mnemonic visuals and game-based tasks. The contribution is an integration-focused system design that orchestrates semantically aligned AI-assisted visuals within interactive study and game flows to maintain pedagogical coherence, rather than proposing new generative algorithms.

2 System Design and Implementation

2.1 Technical Framework

The system is implemented in Unity (2022.3 LTS) and targets Meta Quest 3. The core brain model, sourced from the National Institute of Allergy and Infectious Diseases (CC BY 4.0), provides an anatomically detailed reconstruction of 141 bilateral structures.

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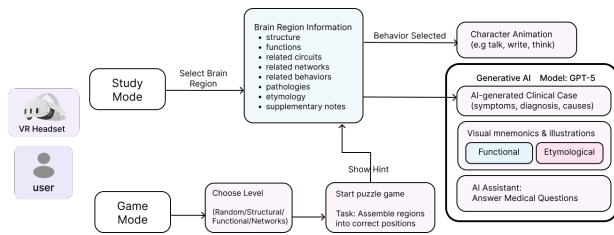


Figure 2: Overall Workflow of the System

A curated CSV knowledge base, reviewed by two medical experts, consolidates neuroanatomy references for each structure, including functions, related networks, pathologies, etymology, and supplementary notes.

For AI-assisted content, OpenAI GPT-5 provides conversational guidance and short, context-aligned case vignettes, while Sora generates mnemonic illustrations. Model choice reflects terminology handling, controllability via structured prompts, and licensing compatibility with educational use. All AI-generated assets are curated for alignment with the knowledge base and presented within interactive Study and Game flows; the timing of asset creation is not emphasized in this work.

A 3D avatar assistant, created with Ready Player Me and animated via Mixamo and DeepMotion, provides in-VR guidance and lightweight demonstrations.

2.2 Interaction Design

The system features two core modes: Study Mode and Game Mode.

2.2.1 Study Mode. In Study Mode, users navigate a high-fidelity 3D brain model and query a curated neuroanatomy knowledge base covering structures, functions, pathways, and related pathologies. Selecting a knowledge entity highlights all corresponding regions, supporting systematic learning; for example, choosing the Default Mode Network (DMN) illuminates its distributed regions and functional relationships.

To transform abstract concepts into tangible and memorable learning experiences, the system integrates three AI-assisted components:

- (1) **Visual Mnemonics.** Curated AI-generated illustrations serve as contextual anchors that bridge abstract neuroanatomy with concrete imagery. For functional understanding, complex processes (e.g., emotion regulation, visual processing) are expressed as intuitive visual metaphors. For terminology retention, etymology fields in the knowledge base link medical terms to linguistic roots—for instance, associating “hippocampus” with a seahorse. These visuals are presented within the interactive study flow and aligned with the curated knowledge base to support initial comprehension and long-term recall.
- (2) **Contextualized Case Vignettes.** Brief, context-aligned clinical vignettes for region-linked conditions illustrate symptoms, mechanisms, and diagnostic reasoning, linking anatomy to clinical application and fostering spatial-clinical associations; terminology is curated for consistency with the knowledge base.

- (3) **Conversational AI Assistant.** An avatar-based assistant provides responsive explanations to user queries and can trigger brief animated demonstrations (e.g., motor actions, speech production) to link anatomical concepts to observable behaviors and reinforce understanding through visual context.

2.2.2 Game Mode. Game Mode is designed to support spatial memory and anatomical localization through a gamified 3D assembly task. Learners select challenges organized by structure, function, network association, or randomized layouts and place regions in their correct locations within a time window.

To support learning rather than guesswork, the system offers knowledge-base-aligned hints (e.g., adjacency or pathway cues) that sustain spatial reasoning while guiding progress. Task framing is progressively structured: higher recent error rates may yield fewer candidate pieces or higher-level hints, whereas lower error rates retain larger sets. These adjustments are lightweight and presentation-oriented rather than algorithmic, intended to sustain engagement while preserving pedagogical coherence.

2.3 Preliminary Assessment

This poster reports a small formative assessment focused on usability and perceived learning support. Seven medical students completed a single 20-minute session that combined the Study and Game modes, followed by a brief post-session questionnaire (Likert items plus free-text). Descriptive observations indicate that participants generally reported clearer spatial relationships and found the AI-assisted mnemonic visuals helpful for recall; the gamified framing was frequently described as engaging. Several remarks noted initial VR acclimation and occasional UI navigation friction. No controlled comparison or inferential statistics were performed, and no quantitative claims are made; these observations informed minor interface refinements and the design of a subsequent controlled evaluation plan.

3 Conclusion and Future Work

This work presents a gamified VR system for neuroanatomy that lets learners examine, disassemble, and reassemble brain structures in an immersive setting with curated, AI-assisted support, designed to aid spatial understanding and retention while maintaining pedagogical coherence.

Future work will conduct systematic, mixed-methods evaluation of educational effectiveness. Quantitative studies will compare outcomes against a control condition, measuring knowledge retention, spatial localization accuracy, and task efficiency; qualitative methods (e.g., interviews and behavioral observations) will capture user experience and inform iterative design. Beyond evaluation, we plan to refine curation and presentation procedures for AI-assisted assets and to explore lightweight personalization—such as adjusting hint granularity or revisit frequency based on recent performance—without emphasizing generation timing or proposing new generative algorithms.

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